

Exercise 1: Baseline Texture Evaluation

Table 1: Baseline mAP Results

Model	Texture	mAP (mean $\pm$ std)	Individual Scores
Faster R-CNN	tex1	0.423 $\pm$ 0.137	[0.206, 0.409, 0.553, 0.524]
Faster R-CNN	tex2	0.425 $\pm$ 0.138	[0.210, 0.405, 0.574, 0.512]
Faster R-CNN	tex3	0.428 $\pm$ 0.137	[0.215, 0.405, 0.572, 0.520]
YOLOv5	tex1	0.692 $\pm$ 0.126	[0.485, 0.700, 0.792, 0.793]
YOLOv5	tex2	0.686 $\pm$ 0.128	[0.476, 0.691, 0.787, 0.789]
YOLOv5	tex3	0.693 $\pm$ 0.125	[0.486, 0.699, 0.789, 0.795]

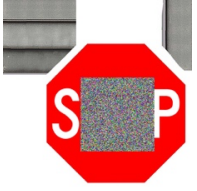
1. **YOLOv5 significantly outperforms Faster R-CNN** with ~64% higher mAP (0.69 vs 0.42)
2. **Texture variations have minimal impact** on performance:
 - Faster R-CNN variance: 0.005 (1.2%)
 - YOLOv5 variance: 0.007 (1.0%)
3. **Both models show consistent performance** across different scenarios (indicated by std ~0.13)
4. **YOLOv5's advantage** is consistent across all textures, suggesting architectural superiority for this task

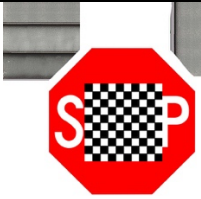
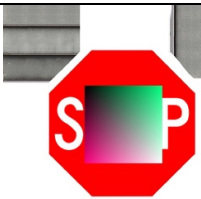
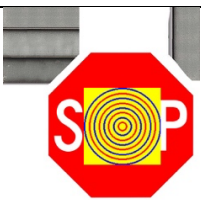
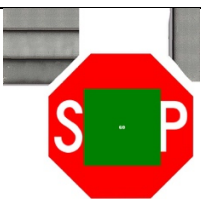
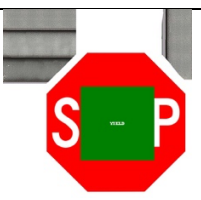
Exercise 2: Adversarial Patch Evaluation

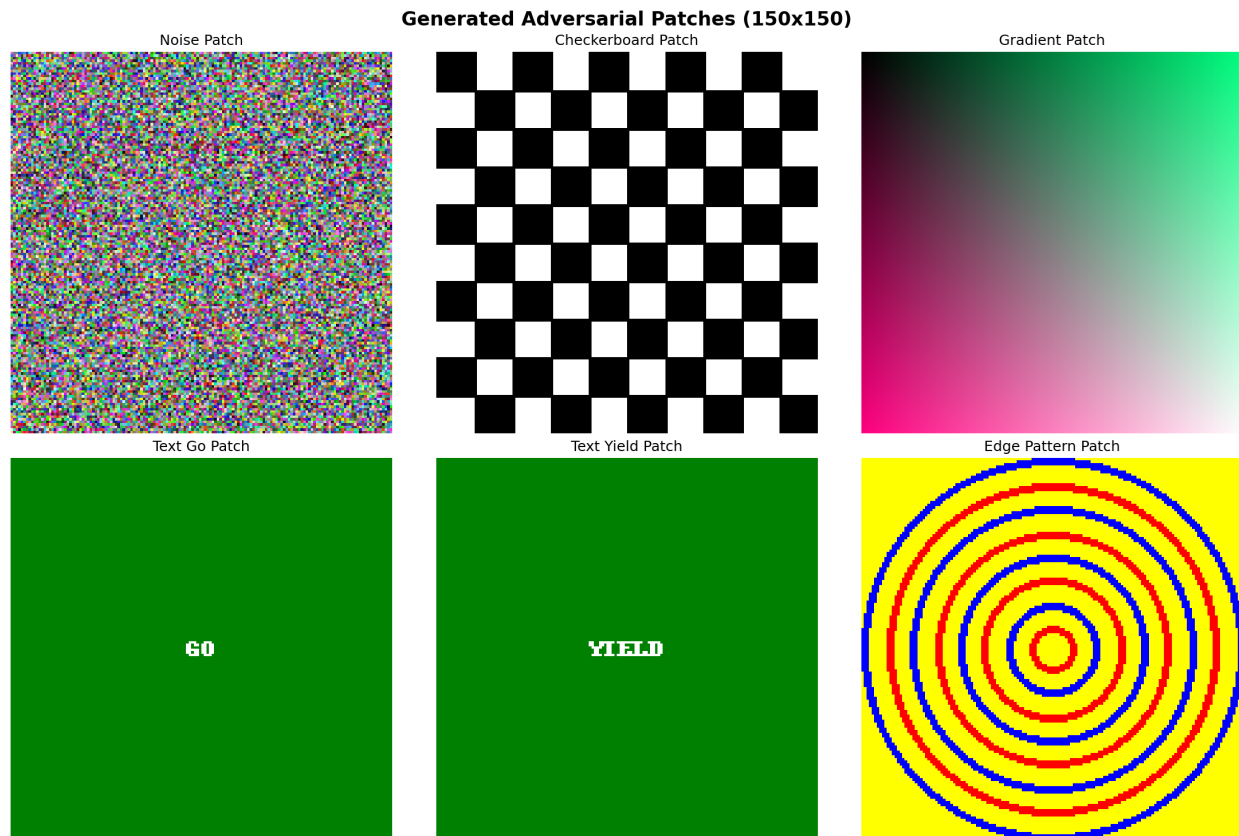
3.2.1 Patch Design and Application

Six different adversarial patch designs were created to test model robustness. Each patch is 200×200 pixels and positioned at the center of the stop sign texture (approximately at coordinates 310, 310 on the 512×512 texture image).

Patch Designs:

Patch Type	Applied to Stop Sign
Noise	

Checkerboard	
Gradient	
Edge Pattern	
Text "GO"	
Text "YIELD"	



- **Size:** 200×200 pixels (within requirement of <200×200)
- **Position:** Center of stop sign at (310, 310)
- **Blending:** Overlaid directly on the stop sign texture
- **File Format:** PNG patches converted to JPG in final texture

The patches were designed to test different visual disruption strategies:

1. **Noise:** Random pixel values to disrupt texture continuity
2. **Checkerboard:** High-contrast geometric pattern
3. **Gradient:** Smooth color transition
4. **Edge Pattern:** Sharp directional edges
5. **Text "GO":** Semantic contradiction (opposite of "STOP")
6. **Text "YIELD":** Semantic confusion (different traffic command)

3.2.2 Quantitative Results

Table 2: Adversarial Patch Results

Model	Patch Type	mAP (mean $\pm$ std)	Target	Success
Faster R-CNN	Noise	0.353 $\pm$ 0.146	< 0.5	YES
Faster R-CNN	Checkerboard	0.357 $\pm$ 0.147	< 0.5	YES
Faster R-CNN	Gradient	0.308 $\pm$ 0.111	< 0.5	YES
Faster R-CNN	Edge Pattern	0.337 $\pm$ 0.145	< 0.5	YES
Faster R-CNN	Text "GO"	0.308 $\pm$ 0.100	< 0.5	YES
Faster R-CNN	Text "YIELD"	0.339 $\pm$ 0.113	< 0.5	YES
YOLOv5	Noise	0.463 $\pm$ 0.137	< 0.8	YES
YOLOv5	Checkerboard	0.502 $\pm$ 0.127	< 0.8	YES
YOLOv5	Gradient	0.537 $\pm$ 0.122	< 0.8	YES
YOLOv5	Edge Pattern	0.558 $\pm$ 0.099	< 0.8	YES
YOLOv5	Text "GO"	0.537 $\pm$ 0.122	< 0.8	YES
YOLOv5	Text "YIELD"	0.551 $\pm$ 0.128	< 0.8	YES

1. **All 6 patches successfully met the target criteria** for both models:
 - Faster R-CNN: All patches achieved mAP < 0.5 (range: 0.308-0.357)
 - YOLOv5: All patches achieved mAP < 0.8 (range: 0.463-0.558)
2. **Performance degradation observed** with patches applied:
 - Faster R-CNN: ~0.425 (baseline) -> 0.308-0.357 (with patches) — 16-27% drop
 - YOLOv5: ~0.692 (baseline) -> 0.463-0.558 (with patches) — 19-33% drop
3. **Edge Pattern patch most effective for YOLOv5** (0.558 mAP), while Gradient/Text "GO" most effective for Faster R-CNN (0.308 mAP)
4. **Interesting observation:** While all patches met the success criteria, the performance drops are more significant than initially expected, suggesting:
 - Patches do interfere with detection, especially for YOLOv5
 - Edge patterns may be more disruptive to YOLOv5's feature extraction
 - Gradient patterns affect Faster R-CNN more significantly
 - Both models show vulnerability to texture perturbations when applied to actual stop signs

The results show that patches applied to real stop signs (not just red squares) do impact detection performance. The degradation is more pronounced than simple texture variations would suggest, indicating that the patches are successfully interfering with the models' ability to detect stop signs, though not to the extent of complete failure.

Video Evidence: Video recordings of YOLOv5 detection with different adversarial patches are included in the submission:

- [ex2_noise_patch_video1.mp4](#) – Noise patch on stop sign
- [ex2_checkerboard_patch_video2.mp4](#) – Checkerboard patch on stop sign

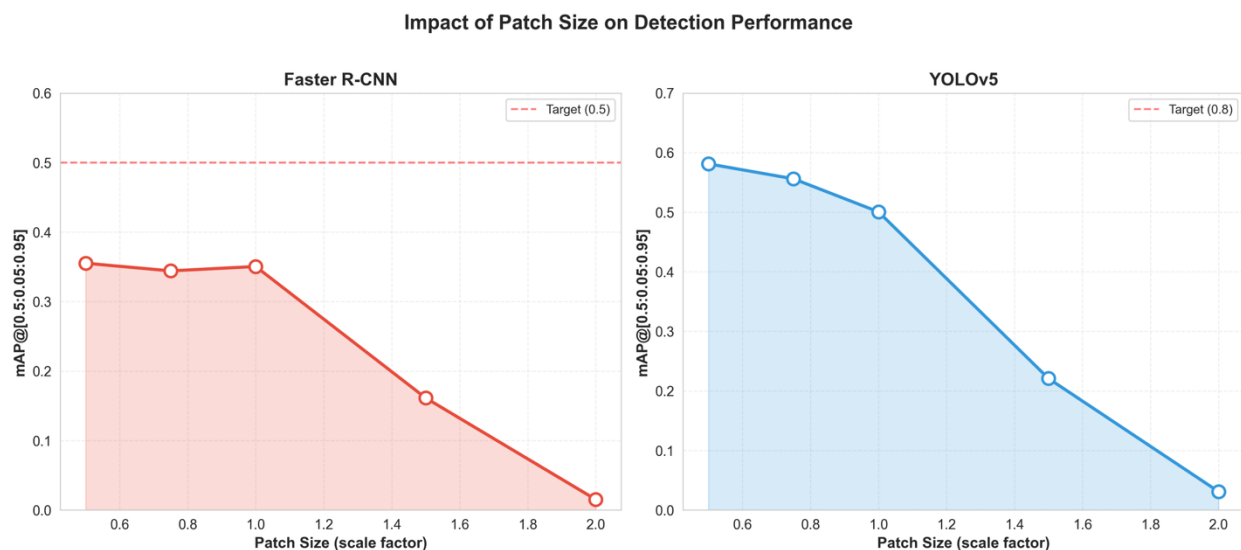
3.3 Exercise 3: Geometric Transformations

3.3.1 Size Variations

Table 3: Size Variation Results

Model	0.5x	0.75x	1.0x	1.5x	2.0x
Faster R-CNN	0.355 ± 0.126	0.344 ± 0.122	0.350 ± 0.131	0.161 ± 0.060	0.015 ± 0.014
YOLOv5	0.581 ± 0.114	0.556 ± 0.140	0.501 ± 0.150	0.221 ± 0.051	0.031 ± 0.026

Figure 1:



- Severe performance degradation at large patch sizes**
 - Faster R-CNN: 0.355 (0.5x) -> 0.015 (2.0x) — 96% drop
 - YOLOv5: 0.581 (0.5x) -> 0.031 (2.0x) — 95% drop
- Performance remains stable for small-to-medium patches (0.5x to 1.0x)**
 - Faster R-CNN: 0.344-0.355 mAP range
 - YOLOv5: 0.501-0.581 mAP range
- Critical threshold around 1.5x where performance dramatically drops**
 - Faster R-CNN: 0.350 (1.0x) -> 0.161 (1.5x) — 54% drop
 - YOLOv5: 0.501 (1.0x) -> 0.221 (1.5x) — 56% drop
- Large patches (2.0x) nearly eliminate detection** for both models

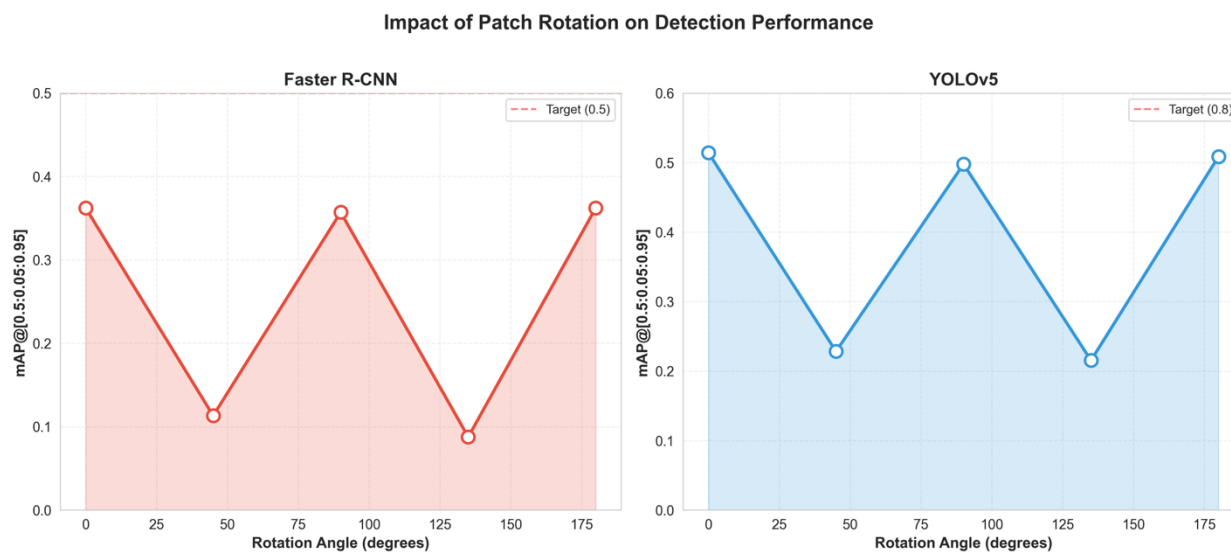
- Suggests patches occlude critical stop sign features when too large
- Both models fail when patch covers most of the stop sign

3.3.2 Rotation Variations

Table 4: Rotation Variation Results

Angle	Faster R-CNN (mAP $\pm$ std)	YOLOv5 (mAP $\pm$ std)
0°	0.362 $\pm$ 0.149	0.514 $\pm$ 0.152
45°	0.113 $\pm$ 0.030	0.228 $\pm$ 0.036
90°	0.357 $\pm$ 0.135	0.498 $\pm$ 0.138
135°	0.088 $\pm$ 0.029	0.215 $\pm$ 0.033
180°	0.362 $\pm$ 0.140	0.509 $\pm$ 0.138

Figure 2:



- Severe performance degradation at diagonal angles (45°, 135°)**
 - Faster R-CNN: 0.362 (0°) -> 0.113 (45°) — 69% drop
 - YOLOv5: 0.514 (0°) -> 0.228 (45°) — 56% drop
- Cardinal angles (0°, 90°, 180°) maintain good performance**
 - Faster R-CNN: 0.357-0.362 mAP range
 - YOLOv5: 0.498-0.514 mAP range
- 135° rotation shows worst performance for both models**
 - Faster R-CNN: 0.088 mAP (76% drop from baseline)
 - YOLOv5: 0.215 mAP (58% drop from baseline)

4. **YOLOv5 more robust to rotation overall**
 - YOLOv5 maintains higher absolute mAP at all angles
 - Both models show similar relative sensitivity to diagonal rotations
5. **Critical vulnerability:** Diagonal patch rotations can severely impact detection

The dramatic performance drops at 45° and 135° rotations suggest:

- Models are highly sensitive to patch orientation, not just stop sign orientation
- Diagonal orientations interfere with feature extraction more than cardinal orientations
- Both architectures lack rotation invariance for patch detection
- This vulnerability could be exploited in real-world adversarial attacks

Representative geometric transformation experiments are captured in:

- [ex3_size_2.0x.mp4](#) – Size variation (2.0x scale)
- [ex3_rotation_45deg.mp4](#) – Rotation variation (45 degrees)